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HPc+DL ORAL (QA)

High Performance Computing (HPC)

1. What is the difference between BFS and DFS?

BFS (Breadth First Search) traverses the graph level by level, visiting all neighboring nodes first. DFS (Depth First Search) traverses deeply into one branch before backtracking. BFS uses a queue while DFS uses a stack or recursion.

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2. Why is BFS more suitable for queue data structure?

BFS follows FIFO (First In First Out) order. The node visited first should be processed first, so queue is the most suitable data structure.

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3. Why is DFS generally implemented using stack or recursion?

DFS follows LIFO (Last In First Out) order. Stack naturally supports this behavior, and recursion internally uses stack memory.

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4. What is parallel computing?

Parallel computing is a technique where multiple computations are executed simultaneously using multiple processors or threads to improve performance and reduce execution time.

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5. What is OpenMP?

OpenMP is an API used for parallel programming in shared memory systems. It provides compiler directives, functions, and environment variables to create multithreaded programs in C, C++, and Fortran.

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6. What is the purpose of #pragma omp parallel?

This directive creates multiple threads and divides the work among them so tasks can execute simultaneously.

7. What is thread synchronization?

Thread synchronization is the coordination between multiple threads to safely access shared resources and avoid data inconsistency.

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8. What problems occur in parallel BFS?

Problems include race conditions, synchronization overhead, and uneven workload distribution among threads.

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9. What is race condition?

A race condition occurs when multiple threads access and modify shared data at the same time, leading to unpredictable results.

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10. How do you avoid race conditions in OpenMP?

Race conditions can be avoided using critical sections, locks, atomic operations, and reduction clauses.

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11. What is speedup in parallel algorithms?

Speedup measures performance improvement and is calculated as:

[
Speedup = \frac{Sequential\ Time}{Parallel\ Time}
]

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12. Difference between sequential and parallel execution?

Sequential execution performs one task at a time, whereas parallel execution performs multiple tasks simultaneously.

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13. Why is graph traversal difficult to parallelize?

Graph nodes are interconnected and dependent on each other, making synchronization difficult during parallel traversal.

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14. What is shared memory architecture?

In shared memory architecture, all processors share a common memory space and communicate through it.

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15. What is load balancing in HPC?

Load balancing means distributing tasks equally among processors so no processor remains idle or overloaded.

Sorting Algorithms

16. Why is Bubble Sort inefficient for large datasets?

Bubble Sort performs repeated comparisons and swaps, making it slow for large datasets due to high time complexity.

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17. What is the time complexity of Bubble Sort?

Worst and average case time complexity is $O(n^2)$.

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18. What is the time complexity of Merge Sort?

Merge Sort has time complexity $O(n \log n)$, making it faster for large datasets.

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19. Why is Merge Sort better for parallelization?

Merge Sort uses divide-and-conquer approach, so subarrays can be processed independently in parallel.

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20. Explain divide and conquer approach.

The problem is divided into smaller subproblems, solved separately, and then combined to get the final result.

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21. How does OpenMP improve sorting performance?

OpenMP divides sorting tasks among multiple threads, reducing execution time.

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22. What is omp for?

omp for distributes loop iterations among multiple threads automatically.

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23. What is chunk scheduling in OpenMP?

Chunk scheduling divides loop iterations into chunks and assigns them to threads.

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24. Difference between static and dynamic scheduling?

- Static scheduling assigns work before execution.
- Dynamic scheduling assigns work during runtime based on availability.

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25. Why is Merge Sort stable?

Merge Sort preserves the relative order of equal elements after sorting.

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26. Define scalability.

Scalability is the ability of a parallel system to improve performance when more processors are added.

Parallel Reduction

27. What is reduction operation?

Reduction combines multiple values into a single value like sum, minimum, or maximum.

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28. Why is reduction useful in parallel computing?

It allows efficient parallel computation of aggregate operations while avoiding race conditions.

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29. Explain OpenMP reduction clause.

The reduction clause creates private copies of variables for each thread and combines results automatically.

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30. How do you calculate average using reduction?

First calculate total sum using reduction, then divide by number of elements.

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31. Difference between reduction and critical section?

Reduction is optimized and faster, while critical section allows only one thread at a time.

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32. What is overhead in parallel computing?

Overhead refers to extra computation time required for thread management and synchronization.

CUDA Programming

33. What is CUDA?

CUDA is NVIDIA's parallel computing platform used for GPU programming.

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34. Difference between CPU and GPU?

CPU has fewer powerful cores for sequential tasks, while GPU has thousands of smaller cores for parallel tasks.

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35. Why are GPUs faster for parallel tasks?

GPUs execute many threads simultaneously, making them suitable for matrix and vector operations.

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36. What is a CUDA kernel?

A CUDA kernel is a function executed on the GPU by multiple threads.

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37. What is thread block in CUDA?

A thread block is a group of threads that can cooperate using shared memory.

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38. What is grid in CUDA?

A grid is a collection of multiple thread blocks.

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39. Explain thread indexing in CUDA.

Thread indexing uniquely identifies each thread using block and thread IDs.

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40. What is global memory in CUDA?

Global memory is GPU memory accessible by all threads but slower than shared memory.

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41. What is shared memory?

Shared memory is fast on-chip memory shared among threads of the same block.

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42. Why is shared memory faster?

Because it is located closer to processing cores.

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43. What is CUDA C?

CUDA C is an extension of C programming language for GPU programming.

44. Explain vector addition using CUDA.

Each GPU thread adds corresponding elements of two vectors in parallel.

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45. How is matrix multiplication performed on GPU?

Each thread calculates one element of the resultant matrix simultaneously.

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46. What is memory transfer between host and device?

Data is copied between CPU memory and GPU memory using cudaMemcpy().

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47. Difference between host and device?

Host refers to CPU and device refers to GPU.

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48. What is __global__ keyword?

It defines a kernel function executed on GPU and callable from CPU.

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49. What is cudaMalloc()?

It allocates memory on GPU device.

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50. What is cudaMemcpy()?

It copies data between host and device memory.

Deep Learning

57. What is linear regression?

Linear regression is a supervised learning algorithm used to predict continuous numerical values using a linear relationship between input and output.

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58. What is the Boston Housing Dataset?

It is a dataset containing housing information used for predicting house prices.

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59. Is Boston Housing a regression or classification problem?

It is a regression problem because output is a continuous value.

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60. What is target variable in Boston dataset?

The target variable is the house price.

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61. What is a Deep Neural Network?

A DNN is a neural network with multiple hidden layers capable of learning complex patterns.

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62. Difference between ANN and DNN?

DNN has many hidden layers, while ANN may contain only one or few hidden layers.

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63. What are epochs?

An epoch is one complete pass of the entire training dataset through the model.

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64. What is batch size?

Batch size is the number of training samples processed before updating weights.

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65. What is loss function?

A loss function measures the difference between actual and predicted output.

66. Which loss function is commonly used for regression?

Mean Squared Error (MSE) is commonly used.

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67. What is Mean Squared Error?

MSE calculates average squared difference between predicted and actual values.

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68. What optimizer did you use?

Adam optimizer because it provides faster convergence and adaptive learning.

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69. What is gradient descent?

Gradient descent is an optimization algorithm used to minimize loss by updating weights.

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70. What is learning rate?

Learning rate controls how much weights are updated during training.

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71. What happens if learning rate is too high?

The model may overshoot the minimum point and fail to converge.

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72. What is overfitting?

Overfitting occurs when model performs well on training data but poorly on unseen data.

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73. How do you prevent overfitting?

Using dropout, regularization, early stopping, and larger datasets.

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74. What is train-test split?

Dataset is divided into training and testing sets to evaluate model performance.

CNN

85. What is CNN?

CNN is a deep learning algorithm mainly used for image processing and classification.

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86. Why are CNNs mainly used for images?

Because CNN automatically extracts spatial and visual features from images.

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87. What is convolution operation?

Applying filters over an image to extract important features.

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88. What is a filter/kernel?

A small matrix used to detect patterns like edges and textures.

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89. What is feature extraction?

The process of identifying important characteristics from input data.

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90. What is pooling?

Pooling reduces feature map size and computation.

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91. Difference between max pooling and average pooling?

Max pooling selects maximum value while average pooling calculates average value.

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92. What is flatten layer?

It converts multidimensional data into one-dimensional vector before dense layer.

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93. Why use ReLU activation?

ReLU improves training speed and reduces vanishing gradient problem.

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94. What is stride?

Stride defines how many pixels the filter moves at a time.

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95. What is padding?

Padding adds extra pixels around image borders to preserve dimensions.

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96. What is overfitting in CNN?

Model memorizes training images instead of learning generalized patterns.

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97. Why is Fashion MNIST used?

Because it is simple and useful for image classification practice.

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98. Difference between CNN and traditional ANN?

CNN automatically extracts image features while ANN requires manual feature extraction.

RNN

99. What is RNN?

RNN is a neural network designed for sequential data processing.

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100. Why are RNNs suitable for time series data?

Because they remember previous inputs using hidden states.

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101. What is hidden state in RNN?

It stores information from previous time steps.

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102. What is sequence data?

Data where order matters, such as text and stock prices.

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103. What is vanishing gradient problem?

Gradients become very small during backpropagation, making learning difficult.

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104. Difference between RNN and LSTM?

LSTM solves long-term dependency problems better than standard RNN.

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105. Why use Google stock price dataset?

It is commonly used for time-series prediction experiments.

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106. What is time series forecasting?

Predicting future values using historical data.

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107. What are timestamps in stock prediction?

They represent date and time information associated with stock values.

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108. Why normalize stock data before training?

Normalization improves training stability and convergence speed.